

Question number	Answer	Additional guidance	Mark
4(a)(i)	swollen / enlarged (hands)	ACCEPT oedema IGNORE other symptoms	(1)

Question number	Answer	Additional guidance	Mark
4(a)(ii)	Any two from: pain / hurts / tender / aches redness / red warmth / heat / increased temperature / hot	IGNORE swelling IGNORE immobility / itching IGNORE fever	(1)

Question number	Answer	Additional guidance	Mark
4(b)(i)	An explanation that includes two of the following points: - because when the virus replicates the {DNA / gene} will be transcribed (1) - and when the RNA is translated the {protein / TNF} will be synthesized (1) - TNF incorporated into capsid when virus is assembled (1)	ACCEPT RNA / mRNA will be made ACCEPT description ACCEPT when new particles are made NB The {gene / DNA} is transcribed and translated = 1 mark if no other mark awarded	(2)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	An explanation that includes two of the following points: - antibody {binds to / neutralises / agglutinates} TNF (1) - therefore will prevent the TNF from binding to the cells (1) - and therefore inflammatory responses will not be triggered (1)	DO NOT ACCEPT antibody binds to cells / antibody destroys TNF IGNORE opsonisation DO NOT ACCEPT antibody binds to receptors (on the cells) ACCEPT inflammation will {not occur / be reduced}	(2)

Question number	Answer	Additional guidance	Mark
4(b)(iii)	An explanation that includes four of the following points: • because (as a result of the TNF antibodies binding to TNF) phagocytosis (by macrophages) will {not happen / be reduced} (1) • therefore {fewer bacteria will be destroyed / bacteria will increase in number} (if less phagocytosis) (1) • credit details of what will not take place if <u>macrophages</u> are impaired (1) • therefore tubercles (more likely to) form (1) • credit example of how TB can cause death (1)	e.g. antigen presentation / activation of T helper cells / humoral immune response e.g. destruction of lung tissue / organ failure / opportunistic infection / pneumonia / HIV / lung damage	(4)

